

Problems Resolution Report

A complete disposition of the findings in *Berry_Problems_Found.pdf*, including the agreed product decisions, the fixes completed in this pass, and the evidence used to verify the current web implementation.

REVIEW STATE	SCOPE
Ready for code review	Web-first Berry platform
All implementation is complete and remains uncommitted.	13 findings reviewed against the user-approved scope.

0

pending in agreed scope

5

implemented or polished

8

verified or accepted

Prepared 4 August 2026

Repository baseline: 43f1e10 | Deliverable: local, gitignored PDF

Executive summary

Every finding has been classified against the current code and the product decisions supplied for this review. The five items that required implementation or polish are complete; the remaining eight are verified as already resolved or explicitly accepted in their current form.

Outcome	Count	Meaning
Implemented in this pass	4	A-2, UI-1, UI-3 and UI-4 received end-to-end code changes.
Polished in this pass	1	A-5 kept the existing context display and removed the unwanted diamond arrow.
Verified resolved	5	A-1, A-3, A-4, A-6 and UI-2 are present and working in the current implementation.
Accepted product decision	3	A-7, UI-5 and UI-6 need no additional web implementation at this time.

What changed

- Project Library upload.** A project menu action now opens a file dropzone, uploads with progress into that project's Library scope, reports per-file success or failure, and can be closed when complete.
- Composer attachments.** Images receive thumbnails; other files remain file pills. Pills size to their content up to a maximum width and truncate safely. Long pasted text now becomes a text attachment with two deliberate interaction modes.
- Model effort safety.** The reasoning picker now derives its choices from model capabilities. GLM-5.2 exposes High and Extra high only, preventing a user from selecting an unsupported level.
- Context polish.** The existing percentage, used tokens and remaining tokens presentation remains the accepted solution. Its tooltip arrow was removed.

Review boundary

No commit or staging operation has been performed. Existing unrelated working-tree changes were preserved. This report and its rendering intermediates are excluded by `.gitignore`.

Finding-by-finding status

The matrix below is the concise handoff list. Detailed reasoning follows.

ID	Reported finding	Status	Current disposition
A-1	Project sessions do not share memory	Verified resolved	Project-scoped memory and knowledge retrieval share the workspace identity across tasks.
A-2	Files are not stored in a project	Implemented	Added Upload to project, modal dropzone, progress, and project Library storage.
A-3	Prompt caching not reaching users	Verified resolved	Cache-read tokens are observable above zero, as confirmed in the current UI.
A-4	Cache misses cannot be diagnosed	Verified resolved	Usage events retain cached-token data and expose request-level accounting.
A-5	No context-window fullness view	Polished	Existing percentage, used and remaining token view accepted; diamond removed.
A-6	Fixed overhead ships every message	Verified resolved	Stable context assembly and cached-prefix accounting are present; accepted as done.
A-7	No plan mode	Accepted	Todo mode covers the current web requirement; dedicated plan mode deferred by decision.
UI-1	No image preview	Implemented	Image-only thumbnails; adaptive pill width and bounded truncation.
UI-2	Composer controls overlap placeholder	Verified resolved	Current composer layout keeps controls and editable text in separate regions.
UI-3	Effort picker shows unsupported levels	Implemented	Capabilities drive the menu; GLM-5.2 offers High and Extra high only.
UI-4	Long pasted text is not collapsed	Implemented	Long and very-long paste modes now use editable text attachments.
UI-5	No in-session usage or cost view	Accepted	Current context-window popover is sufficient; no separate usage surface requested.
UI-6	No search in related conversations	Accepted	Typing after the conversation trigger filters indexed matches; explanation is sufficient.

Detailed findings: architecture and behavior

A-1 Shared project memory

Verified resolved

Reported issue. Sessions created inside the same project were reported as isolated and unable to use project knowledge or memory created by another session.

Disposition. The current memory pipeline supports project scope keyed by workspace, and knowledge retrieval is constrained to the same workspace while remaining available across tasks in it. No new code was required in this pass.

Evidence. `apps/api/src/memory/memory.service.ts`; `apps/api/src/memory/memory.repository.ts`; `apps/api/src/knowledge/knowledge.service.ts`

A-2 Upload files directly to a project

Implemented

Reported issue. Files could be attached to one message but there was no direct project action that stored files for later use from the Library.

Disposition. The project three-dot menu now includes **Upload to project**. It opens a compact modal with browse and drag-and-drop input, accepts multiple files, reports upload progress and final status per file, and stores successful files with project visibility in the selected workspace. Closing the modal returns the user to the current conversation.

Evidence. `apps/web/src/components/projects/project-upload-dialog.tsx`; `apps/web/src/components/shell/web-sidebar.tsx`; `apps/web/src/components/app-shell.tsx`

Scope note. The agreed implementation is project Library upload. Version-history and replace-file workflows from the older report are outside this pass. Demo mode intentionally reports that a connected Berry deployment is required.

A-3 Prompt-cache benefit reaches the user

Verified resolved

Reported issue. The old version showed no visible evidence that prefix caching was producing a cache hit.

Disposition. The current usage presentation reports cache-read tokens, and the supplied observation shows a value greater than zero. That is direct evidence that cached input is being used. No further implementation was requested.

Evidence. Current context/usage UI plus cached-token fields in worker usage rollups.

A-4 Cache diagnostics

Verified resolved

Reported issue. The older implementation did not expose enough accounting to distinguish cached and uncached prompt input.

Disposition. The current usage model retains cached-token counts in request and rollup data. Combined with the visible cache-read value, operators can determine whether a request received cache reuse rather than inferring it from total input alone.

Evidence. `apps/worker/src/usage-rollups.ts` and related tests; API usage-event accounting.

Detailed findings: context and execution

A-5 Context-window fullness

Polished

Reported issue. Users needed to see how much context was used, the total window, and what remained.

Disposition. The existing composer popover already shows percentage used, percentage left, used versus total tokens, and tokens remaining. That is the accepted scope; estimated turns remaining is intentionally omitted. This pass removes the distracting white diamond beneath the popover while preserving keyboard and pointer behavior.

Evidence. packages/desktop-ui/src/components/ui/tooltip.tsx; apps/web/src/components/tasks/web-composer.tsx

A-6 Repeated fixed prompt overhead

Verified resolved

Reported issue. The original report identified stable instructions and project context being resent as uncached input on every turn.

Disposition. The present request pipeline separates stable context construction from turn-specific content and records cache-read tokens. The user accepted this item as complete, so no code change was made in this pass.

Evidence. Worker turn assembly, prompt-cache accounting, and the confirmed non-zero cache-read metric.

A-7 Plan mode

Accepted

Reported issue. The report asked for a dedicated planning state before task execution.

Disposition. Todo mode already provides the visible task decomposition needed for the current web product. A separate plan mode is not required now. This is a deliberate product decision rather than an unresolved defect.

Evidence. Current Todo workflow; user-approved scope decision dated 4 August 2026.

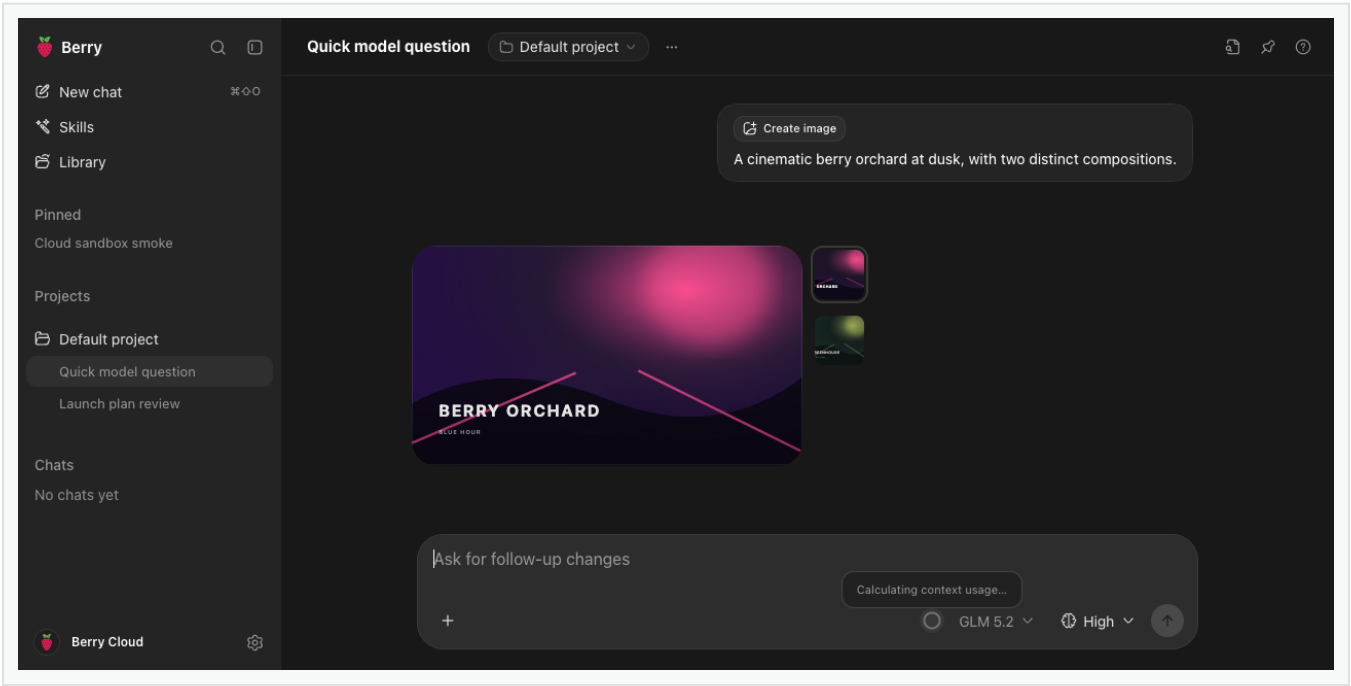


Figure 1. Accepted context-window presentation after removal of the tooltip diamond. Percentage, used tokens and remaining tokens remain visible.

Detailed findings: composer and model controls

UI-1 Image previews and attachment sizing

Implemented

Reported issue. Attachments did not provide an image preview, and a fixed-width pill could waste space or allow long names to dominate the composer.

Disposition. Image MIME types now render a thumbnail from the stored preview URL (or the local demo data URL). Non-image files remain icon-based pills. The pill sizes to its content, is capped at 360 pixels, and truncates its title beyond that boundary. The remove control remains available without expanding the pill.

Evidence. apps/web/src/components/tasks/composer-attachment-pill.tsx; apps/web/src/components/tasks/web-composer.tsx; apps/web/src/styles.css; packages/shared/src/index.ts

UI-2 Composer control overlap

Verified resolved

Reported issue. The previous composer allowed controls to compete with placeholder or prompt text.

Disposition. The current Lexical editor and action controls occupy separate layout regions with reserved spacing. Browser verification at the target desktop viewport showed no overlap in the empty, attachment, and populated states.

Evidence. apps/web/src/components/tasks/web-composer.tsx; browser verification at 1280 x 633.

UI-3 Model-compatible reasoning levels

Implemented

Reported issue. The effort selector could offer levels that the active model did not support. The report specifically noted GLM-5.2 should expose only high and extra-high choices.

Disposition. Reasoning capabilities can now declare an explicit effort list. The composer derives its menu from that list, supports the new xhigh value end to end, and automatically moves away from a stale unsupported selection when the model changes. GLM-5.2 is configured for High and Extra high only.

Evidence. packages/shared/src/index.ts; packages/router-client/src/index.ts; apps/api/src/http/agent-api.controller.ts; apps/worker/src/turn-runner.ts; apps/web/src/lib/env.server.ts; apps/web/src/components/tasks/web-composer.tsx

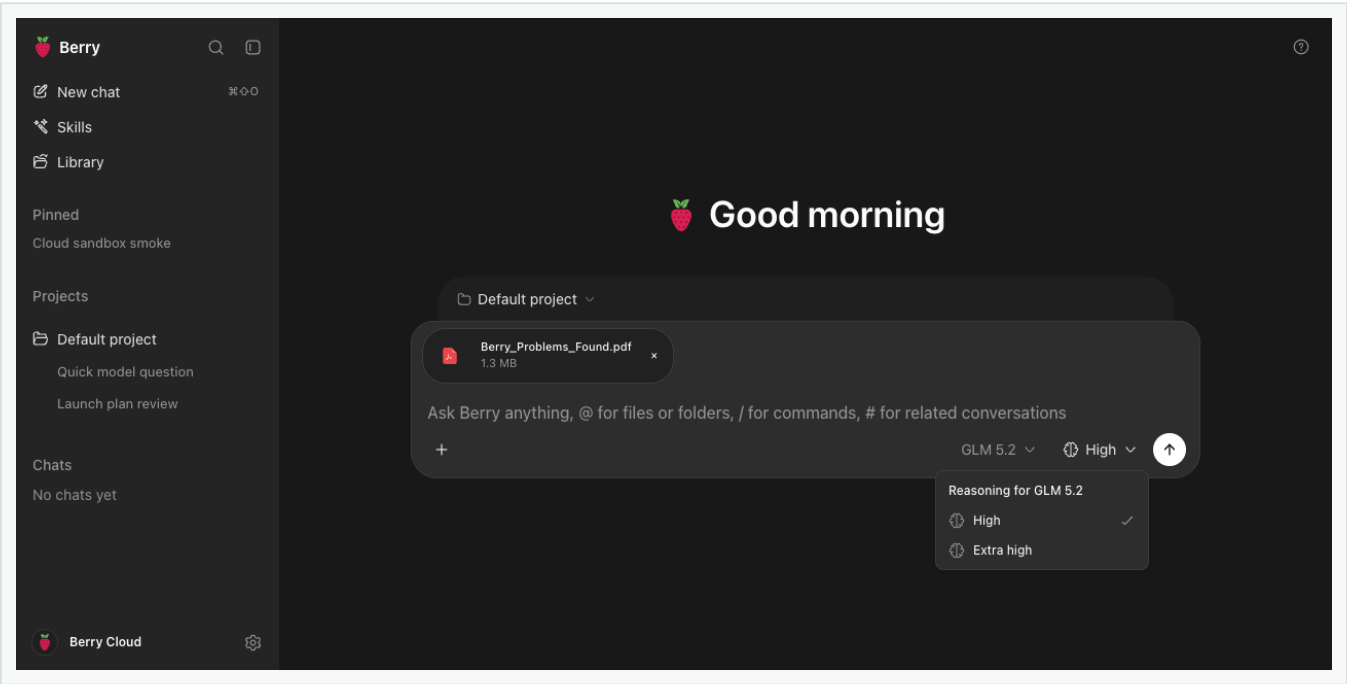


Figure 2. GLM-5.2 presents only High and Extra high, matching the model capability declaration.

Detailed finding: two-mode pasted text

UI-4 Collapse and edit long pasted text

Implemented

Reported issue. Large pasted content previously flooded the prompt field instead of becoming a manageable attachment.

Disposition. Short pastes remain ordinary editor text. From 2,000 through 40,000 characters, a paste becomes **Pasted text.txt** with a title derived from the first non-empty line and a **Show in text field** action. Activating it restores the content to the editor. Above 40,000 characters, the attachment is marked too long for the prompt field; activating it opens an editable text-file modal. Saving creates a replacement text attachment and preserves the normal submit flow.

Evidence. apps/web/src/components/tasks/web-composer.tsx; apps/web/src/components/tasks/composer-attachment-pill.tsx; apps/web/src/components/tasks/pasted-text-editor-dialog.tsx; apps/web/src/components/tasks/web-composer.test.ts

Scope note. The 2,000 and 40,000 character boundaries are explicit implementation constants and have boundary tests.

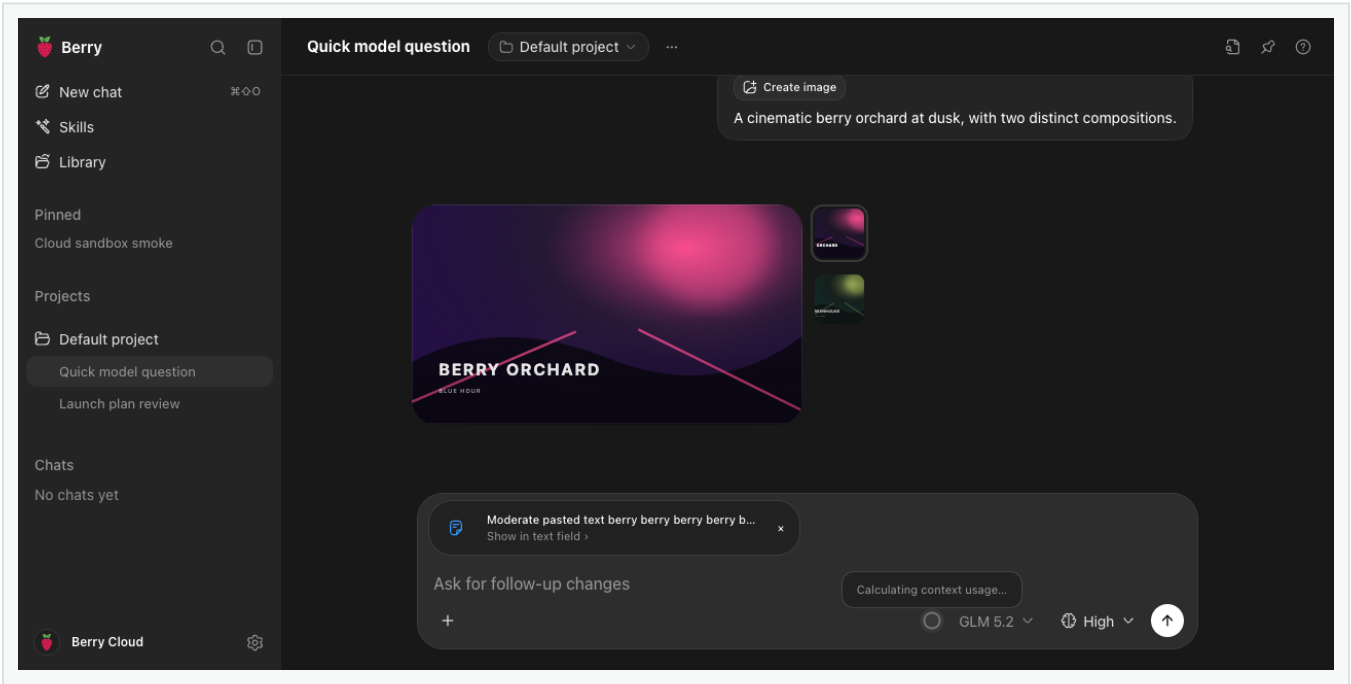


Figure 3. Long-but-manageable paste becomes a compact text attachment with Show in text field.

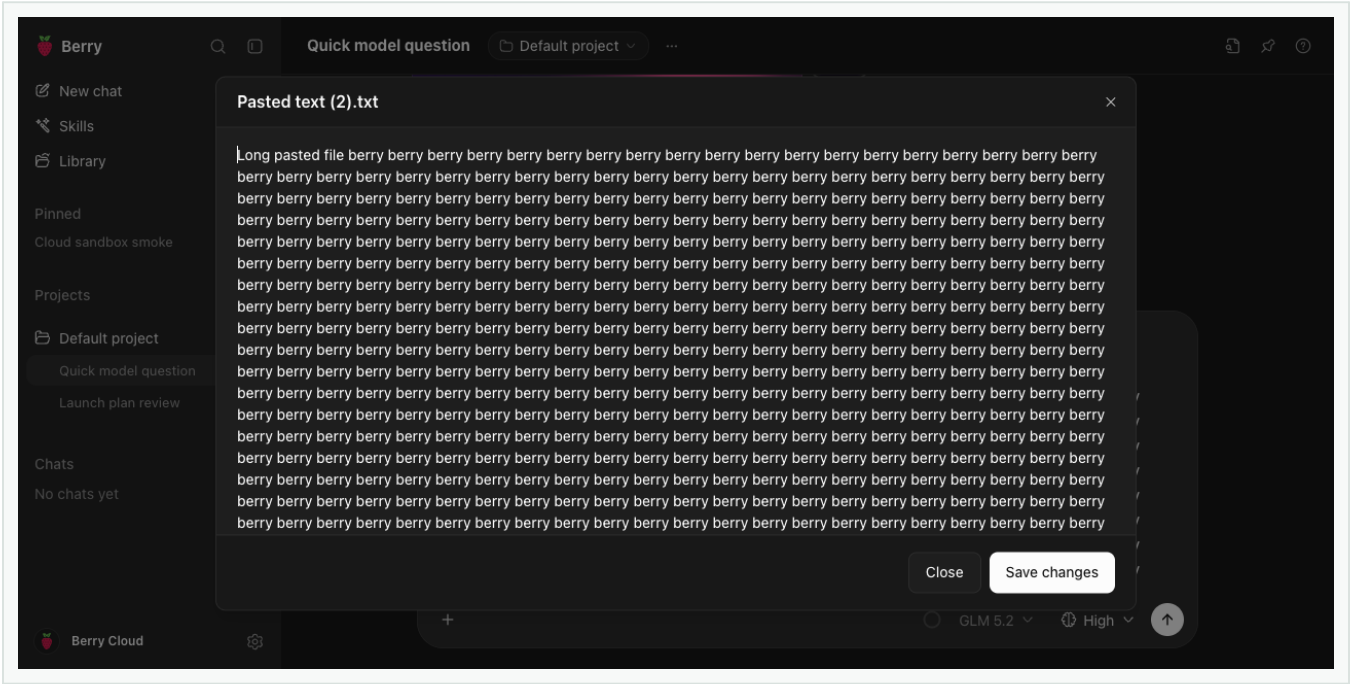


Figure 4. A very long paste opens as an editable text file with Close and Save changes actions.

Detailed findings: accepted current interactions

UI-5 In-session usage view

Accepted

Reported issue. The original report requested a dedicated command or popover for token and cost usage.

Disposition. The existing context-window popover is sufficient for the present web scope. It exposes the context information needed during composition, and no separate cost or slash-command surface is requested now.

Evidence. Current context-window popover; user-approved scope decision dated 4 August 2026.

UI-6 Search related conversations

Accepted

Reported issue. The older report interpreted the related-conversation picker as lacking search.

Disposition. The current mention interaction uses the typed characters after its trigger as an inline query, ranks starts-with and contains matches, and limits the menu to the best results. A separate search input would duplicate that interaction, so explanation of the existing behavior is sufficient.

Evidence. `apps/web/src/lib/mentions.ts`; `apps/web/src/components/mention-menu.tsx`

Interaction details that were intentionally preserved

- Image preview is image-only; PDFs and other document types remain file pills.
- A direct project upload targets the selected project even if another conversation is open.
- Upload modal closure is disabled while files are actively transferring, preventing accidental interruption.
- Pasted text remains editable in both long-text modes.
- Reduced-motion and existing Berry theme variables remain in effect.

Project upload: visual proof

The new action follows a direct, bounded flow: project menu to scoped upload dialog to project-visible Library file.

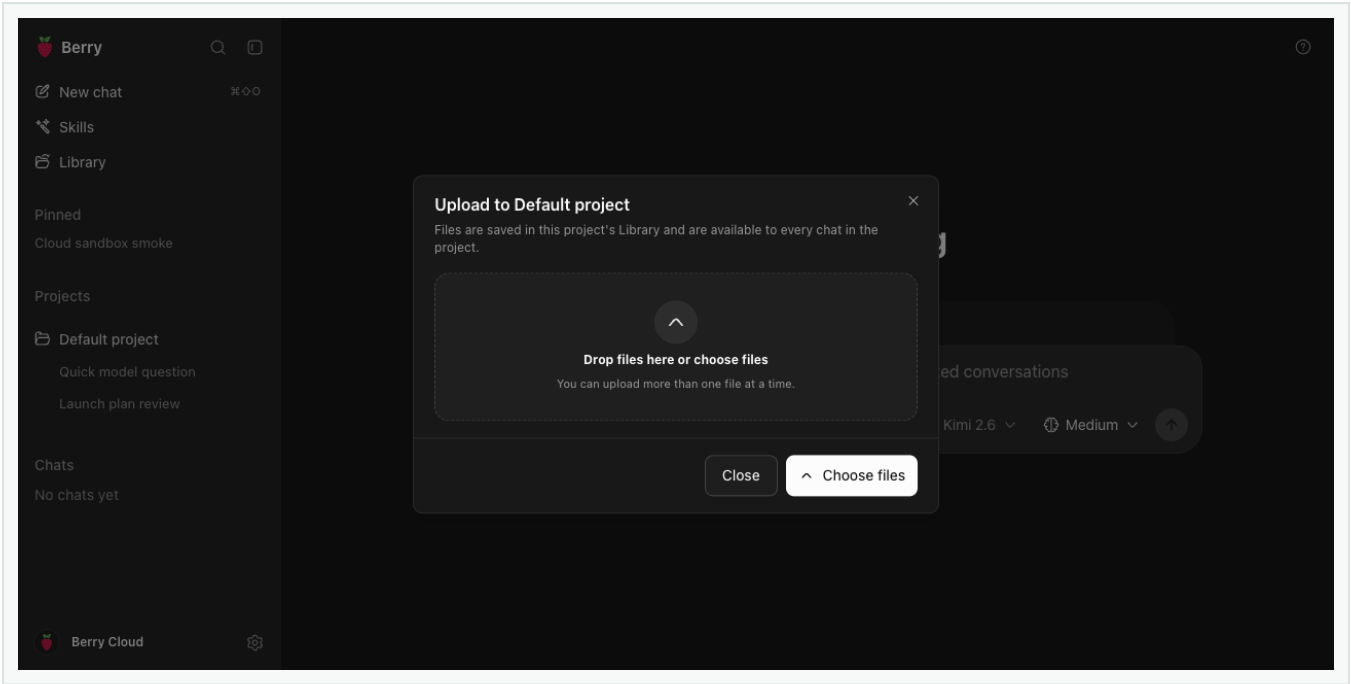


Figure 5. Upload to project opens a scoped modal with drag-and-drop, browse, and multi-file support.

The dialog names the target project before transfer, explains that the result is available to every chat in that project, and supports multiple files. Each upload row then exposes progress and its final success or error state.

Visual verification

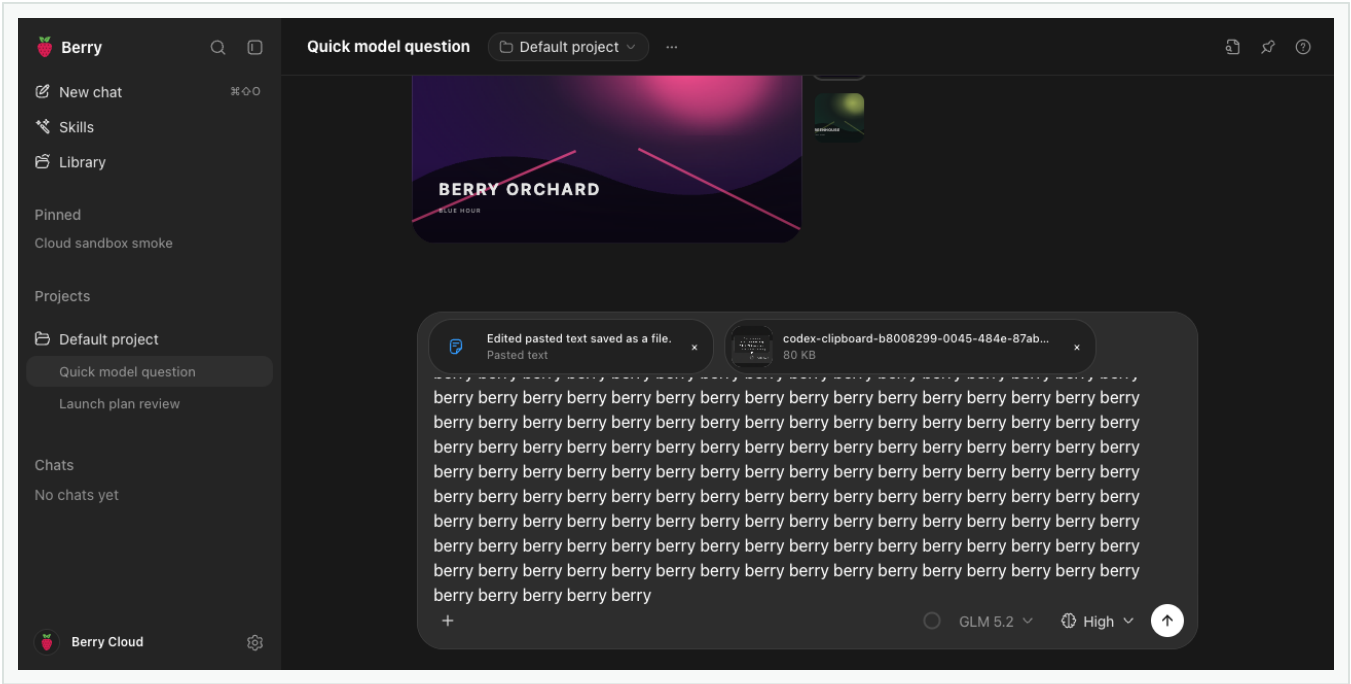


Figure 6. Image attachments display a thumbnail while retaining a compact, bounded attachment pill.

Observed browser behaviors

- Project upload.** The menu action opens the correct project name; the demo-only upload attempt produces an explicit connected-deployment error rather than pretending the file was stored.
- Moderate paste.** The card can restore its content into the Lexical editor and removes the attachment after restoration.
- Very long paste.** The card opens the editor modal, the text area is editable, and Save changes replaces the attachment.
- Reasoning menu.** GLM-5.2 offers exactly High and Extra high.
- Context popover.** Token statistics remain present and the white diamond is absent.
- Runtime health.** Browser error collection was empty. The console contained only development hot-module-reload notices, with no runtime exceptions.

Engineering verification

The checks below were run after the final lazy-loading change that brought the production bundle within its enforced budget.

Check	Evidence	Result
Web typecheck	pnpm --filter @berry/web... typecheck	Passed
API typecheck	pnpm --filter @berry/api... typecheck	Passed
Worker typecheck	pnpm --filter @berry/worker... typecheck	Passed
Web tests	19 files / 65 tests	Passed
Shared tests	4 files / 87 tests	Passed
Router client tests	4 files / 56 tests	Passed
API tests	32 files / 155 tests	Passed
Worker tests	30 files / 128 tests	Passed
Web production build	Vite build plus bundle guard	Passed
Browser flow checks	Project upload, image, two paste modes, reasoning and context	Passed

Bundle result

Final root JavaScript: **1,481,937 bytes raw** and **445,477 bytes gzip**, below the enforced budgets of 1,510,000 and 453,000 bytes. The project upload dialog, pasted-text editor, and attachment pill are lazy-loaded so the new behavior does not penalize initial navigation.

Protocol consistency

The shared attachment schema and model-capability schema changed, so protocol documentation was regenerated. Shared, router-client, API and worker tests then passed against the final schema.

Review checklist

1. Review the uncommitted diff for the project upload workflow.
2. Review the composer thresholds and editing behavior.
3. Review model capability declarations, especially GLM-5.2 high/xhigh.
4. Confirm the PDF wording and accepted product decisions.
5. Commit only after explicit approval.

FINAL STATUS

All 13 findings are resolved, verified, or accepted within the agreed scope.

Zero pending implementation items. Code and report are ready for review and remain uncommitted.